What makes motion hard to stop?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare momentum changes in a controlled stopping situation.

SUCCESS CRITERIA - I can...

1. Compare mass and velocity while controlling other variables.
2. Use momentum change to compare stopping force for equal times.
3. Avoid claiming that momentum alone determines damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

An ideal asteroid orbit has nearest distance 3 AU and farthest distance 6 AU. Where is it fastest? Explain with the equal-area law.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p = mv$; $F_{avg} = \Delta p/\Delta t$.
Momentum p includes direction. For a fixed stopping time, greater momentum change needs greater average net force.

Worked reasoning: A 2 kg cart and a 4 kg cart move right at 3 m/s. Their momenta mv are 6 and 12 kg·m/s. Stopping the second requires twice the momentum change.

Your turn: Draw the velocity and momentum arrows for both carts. Explain which pair has equal lengths.

Marzano check - circle: ① ② ③

03 Look and explain

Carts A and B both move right at 2 m/s. A has mass 1 kg; B has mass 3 kg. Which has more momentum? State the controlled quantity.



A: 1 kg; 2 m/s B: 3 kg; 2 m/s

Marzano check - circle: ① ② ③

04 Build the solution

Both carts stop in 3 s. Use force size=momentum change size÷time. Find each average stopping force size.

Marzano check - circle: ① ② ③

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05 Use the idea

Two objects have equal momentum but strike different materials over different times. Can equal momentum alone establish equal damage? Explain.

Marzano check - circle: 1 2 3

06 Choose a method

A supplied gravity model has $Gm_1m_2=144$ N·m² and separation 6 m. Find the force and its direction.

Marzano check - circle: 1 2 3

07 Exit check

A 2 kg cart moving at 3 m/s stops in 3 s. Find momentum change size and stopping force size.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Find a lighter, faster cart with the same momentum as a 4 kg cart at 3 m/s. Explain why equal momentum does not mean equal mass or speed.

Marzano check - circle: 1 2 3